MODELLING CONVENTIONS

Harborview Securities LLC - Leveraged Finance and Acquisition Finance

Purpose: to standardise day-count, rounding and definitional conventions across deal models so that two analysts modelling the same transaction produce the same figures.

1. Day-count conventions are instrument-specific

A single shared day-count constant applied across a model is the most common source of reconciliation differences. Each instrument carries its own convention and the model must hold them separately.

Instrument	Convention	Note
US corporate note	30/360 Bond Basis	Accrued interest and stub periods
Municipal / industrial revenue bond	30/360 Bond Basis	As above
US Treasury benchmark	Actual/Actual (ICMA)	Never share the corporate constant
Revolver / term loan	Actual/360	Per the credit agreement

Where a model discounts a corporate cash flow against a Treasury benchmark, the two legs use different conventions and the model must not collapse them.

2. Rounding

Round only at declared output boundaries, never mid-formula. Currency to the cent, rates to four decimal places, leverage ratios to four decimal places, spreads to one tenth of a basis point.

Where a figure feeds a threshold test, carry it unrounded into the comparison and round only for presentation. Rounding before a covenant test can change the outcome. The same discipline applies to a leverage-ratio numerator and denominator individually - round the ratio itself once computed, never round each of the numerator and denominator first and then divide, since the two roundings do not commute and a model that rounds early can cross a covenant threshold that the unrounded arithmetic does not.

3. Which definition governs which purpose

Several measures in a leveraged acquisition carry more than one definition. The model must label each instance with the definition used and the document that supplies it. This is not a formality: two definitions of the same-named measure can diverge precisely where the divergence matters, and a model that silently reuses one definition for every purpose has not modelled the transaction, it has modelled a simplification of it.

Measure	Purpose	Governing document
Consolidated EBITDA	Covenant compliance	Credit agreement
Adjusted EBITDA	Diligence presentation	Quality-of-earnings report
Earn-out EBITDA	Earn-out measurement	Purchase agreement earn-out schedule
Net debt	Purchase price adjustment	Purchase agreement
Net debt	Leverage covenant	Credit agreement

A model cell that computes any of the measures above must cite, in its own label or in an adjacent documentation cell, which governing document's definition it applies. A cell labelled simply "EBITDA" with no further qualification is not traceable, and a reviewer cannot tell which of the several defined concepts above it purports to compute.

4. Interpolation on the Treasury curve

The published par yield curve does not carry every tenor. Where a remaining term falls between two published tenors, interpolate linearly on remaining term between the two bracketing tenors.

Do not substitute the next longer tenor, and do not assume the curve rises monotonically. Published curves are frequently inverted over part of their range, and a walk-to-the-next-tenor shortcut returns the wrong rate whenever it is.

5. Measurement periods

Model periods should follow the buyer's fiscal calendar rather than calendar quarters, so that reported results and modelled periods reconcile without adjustment.

For earn-out modelling the team's working convention has been to treat Period 1 as commencing on the Closing Date, which keeps the model's period boundaries aligned with the funds flow and avoids a

partial first period.
Where the transaction documents define the measurement period expressly, the documents govern and this convention is set aside. The modeller should confirm the definition rather than carry the convention forward.

6. Spread measures

Three spread measures appear in credit work and they are not interchangeable. The G-spread is measured against the interpolated government curve. The I-spread is measured against the swap curve. The option-adjusted spread removes the value of any embedded option from the comparison.
For a callable instrument the option-adjusted spread is the only measure that permits comparison against a non-callable instrument, because the holder of a callable bond has implicitly sold an option and a spread measure that ignores that option overstates the instrument's value. Report the G-spread only alongside the option-adjusted spread, never in place of it.
Where a comparison is drawn against a tax-exempt instrument, convert to a taxable-equivalent basis before comparing and state the marginal rate assumed.
Both spread measures above are computed off the same yield leg, and that yield leg is itself a definition the model must get right before either spread is meaningful. Yield to worst is the minimum of the yields to every scheduled call date AND the yield to legal maturity - never the call schedule alone. A callable note's step-down schedule front-loads its call premiums, and for a note trading below every call price the minimum-yield date can fall at legal maturity itself, after the last scheduled call. A search that stops at the call schedule and never tests maturity as a candidate will return a higher yield than the true worst case and understate the spread that follows from it. This yield-to-worst convention governs every relative-value comparison in this deal team's models; a model that reports a single yield without applying it has not computed yield to worst at all.

7. Expected credit loss

The measurement basis depends on the classification of the instrument, and the two bases are structurally different tests rather than variants of one calculation.
For an instrument held to maturity the allowance is a lifetime expected loss measured on amortised cost, built from probability of default, loss given default and exposure at default, with no floor at fair value.
For an available-for-sale instrument only the credit-related component is recognised as an allowance, and that allowance is limited to the amount by which fair value is below amortised cost. The remainder of any decline is not an allowance.
Determine the classification from the issuer's own accounting policy note before selecting a basis. Applying the held-to-maturity basis to an available-for-sale position, or the reverse, produces a materially different figure.

8. Purchase price mechanics

Where a transaction uses completion accounts, the price is re-measured at closing and the model must carry the true-up. Where it uses a locked-box mechanism, the price is fixed off a historical balance sheet and the seller is compensated by a value accrual for the intervening period.
The two mechanisms produce different equity prices from the same enterprise value. A model must not carry provisions from both, and where the diligence record contains a precedent document using the other mechanism, the executed agreement governs.
Where a working-capital collar or dead-band applies under a completion-accounts mechanism, once breached the adjustment to the purchase price is full-difference, not limited to the excess over the collar. A model that adjusts only the excess above the collar once the dead-band is breached is applying the wrong mechanic, and the error compounds because it understates the true-up in exactly the direction that favours whichever party the model happened to be built for.

9. Documentation

Every input cell traces to a source document and a section. Live-link market data to its staged extract rather than typing a value. A typed rate cannot be reconciled and will not survive review.
Where a model cell is expected to compute, it must contain a formula. A constant that happens to equal the right answer is not a calculation and will be treated as an error in review, because it cannot respond to a change in its inputs.

10. Earn-out discounting convention

Two distinct discount rates apply to the earn-out at different points in its life, and the two are not interchangeable. While the payout remains contingent on achieving the applicable EBITDA and revenue thresholds, apply the operational discount rate, which reflects the target's own operational and business risk over the period still to run.
Once a measurement period's Achievement is finalized and the resulting payment is fixed and due, apply the buyer-credit discount rate instead. At that point the operational risk has already resolved one way or the other, and the only remaining risk is the buyer's own credit quality as the obligor on a payment that is now certain in amount.
Discounting a still-contingent payment at the buyer-credit rate understates the risk that the thresholds are never reached; discounting a fixed-and-due payment at the operational rate overstates a risk that has already resolved. Either error misprices the earn-out. Confirm which measurement period each payment corresponds to, and whether that period's Achievement has been finalized, before selecting the applicable discount rate - the choice is not a matter of house convention and does not default to either rate.

11. Funds-flow balancing convention

A closing funds-flow memorandum states more Sources-and-Uses lines than the transaction's other governing documents fix independently, and exactly one line is the model's free variable. That line is resized to make total Sources equal total Uses once every other line, including any redemption cost triggered under the notes' own change-of-control provisions, is known.
Every other Sources-and-Uses line is treated as fixed once populated from its own governing document - the purchase price from the purchase agreement, the revolver draw from the credit agreement's own borrowing base, and so forth. Only the one declared free-variable line absorbs the balancing adjustment; a model that instead resizes several lines at once to force balance has lost the ability to explain which number moved and why.

12. Definitions hierarchy when governing documents conflict

Where two governing documents each state a rule for the same mechanic - a measurement period's start date, a spread convention, a classification test - the model must apply the rule of the document that actually governs the purpose at hand, never the rule of whichever document the modeller consulted first or found easier to read.
A precedent document, a superseded draft, or a working convention memorandum such as this one may describe a rule that is a reasonable simplification, a historical practice, or a convenient default - none of which is the same thing as the governing rule. Before relying on any convention stated here, confirm against the transaction's own executed documents that the convention has not been supplanted for the specific purpose in question.

13. Independent recalculation versus counterparty-supplied schedules

A schedule supplied by a counterparty, an advisor, or management - a synergy calculation, an amortization table, a covenant compliance worksheet - may be used as a starting point, but its arithmetic should be independently recalculated from the same governing inputs before the model relies on its output.
Agreement between the counterparty-supplied figure and the model's own independent recalculation increases confidence in both; disagreement is itself informative and should be resolved before either figure is used, never averaged or otherwise blended away.

14. Comparable company and comparable bond selection conventions

A relative-value comparison is only as good as the comparator set it draws on. Screen candidates on the dimensions that actually drive spread and multiple - industry, size, leverage, rating, and structural seniority - before screening on convenience, such as which issuers happen to have the most recently published financial statements.

Document why each excluded near-comparator was excluded, not only why each included comparator was included. A comparator set assembled only by including convenient matches, with no record of what was screened out and why, cannot be audited by a reviewer who was not in the room when the set was built.

Where a comparable bond trades at a size below the standard reporting threshold, flag the observation rather than silently treating it as equivalent to a standard-size print; a thin, capped, or otherwise size-flagged print carries less information than an ordinary-size trade and should be weighted accordingly in any average or median the model computes across the comparator set.

15. Leverage ratio numerator and denominator conventions

State explicitly, for every leverage ratio the model computes, which debt instruments sit in the numerator and which EBITDA definition sits in the denominator. A ratio labelled only "leverage" with no further specification invites a reviewer to assume the textbook definition, which is frequently not the definition the governing covenant actually tests.

Where the credit agreement computes leverage on a net basis (debt net of unrestricted cash) and a rating agency or an internal credit memorandum computes it on a gross basis, carry both as separate, separately labelled lines rather than converting one into the other inside a single cell whose formula a later reviewer cannot easily unwind.

Confirm, for every reporting period the model covers, whether the leverage calculation is measured as of period end or on an average basis over the period; the two bases can diverge materially for a business with meaningful intra-period seasonality in either revenue or working capital.

16. Covenant compliance certificate cross-check conventions

Before relying on a covenant compliance certificate's own stated ratio, independently recalculate the ratio from the certificate's own underlying components rather than accepting the certificate's bottom-line figure at face value. A certificate is a management representation like any other schedule discussed elsewhere in this memorandum, and the same corroboration discipline applies to it.

Where the certificate's own addback taxonomy differs from the diligence team's own presentation of the same period - for example, because a run-rate item is classified differently for covenant purposes than for diligence purposes - document the difference explicitly rather than reconciling it away inside the model. The difference is itself information a lender or a rating analyst would want to see.

17. Presentation conventions for board and lender materials

Every schedule presented to the board or to a lending syndicate should carry a visible source citation for its own headline figures, in the same style this memorandum's own Section 9 requires of every model input cell. A polished presentation slide with no visible sourcing is not more credible than a working schedule with sourcing; it is only harder to audit.

Where a board or lender presentation rounds a figure for readability, state the rounding convention applied and retain the unrounded figure in the supporting schedule, consistent with Section 2 above. Never let a presentation-layer rounding choice silently become the figure a later reader treats as authoritative.

18. Version control and model change-log conventions

Every model that is circulated outside the immediate deal team carries a version identifier and a change log recording, for each new version, what changed and why. A model that is silently updated in place, with the prior version overwritten and no record of the prior figures, cannot support a later question about why a figure moved between two circulated drafts.

Where a change touches a formula that a covenant test, an earn-out calculation, or a make-whole calculation depends on, the change log entry should identify every downstream cell the change affects, not only the cell that was directly edited. A spreadsheet's own dependency chain is not always visible to a reviewer who did not build the model, and the change log is what makes that chain auditable after the fact.

Reconcile every circulated version against the immediately prior version before sending it forward, specifically on the headline figures a recipient is likely to quote from memory - the enterprise value, the leverage ratio, the earn-out payout range. A silent change to one of these figures, even if fully intentional and fully correct, causes real confusion if the recipient is not told it happened.

19. Independent price and market-data verification conventions

Every market-data input - a Treasury curve point, a comparable bond's quoted price, a swap rate - should be traceable to a staged extract from the source this memorandum's own documentation section requires, and that extract's own retrieval date should be visible alongside the figure. A market input with no visible retrieval date invites a reader to assume it is current when it may not be.

Where two staged extracts of the same nominally identical series disagree - for example, because one was retrieved before, and one after, a same-day revision to the published series - resolve the disagreement explicitly by retrieval date and document which vintage the model uses, rather than picking whichever value happens to support the desired conclusion.

For a thinly traded instrument, prefer a size-qualified print or a matrix-priced estimate that discloses its own qualification over an unqualified quoted price that does not disclose whether it reflects an executed trade or an indicative dealer quote; the two carry materially different reliability and should never be presented identically in a model or a memorandum.

20. Sign convention and presentation of negative figures

Present every negative figure in parentheses, never with a leading minus sign, consistent with the accounting-alignment convention this firm's other work product follows. A model that mixes minus-sign negatives in one schedule with parenthetical negatives in another invites a misread the first time a reviewer moves quickly between the two.

Adopt a single sign convention for each schedule as a whole - for example, outflows negative and inflows positive in a funds-flow schedule - and state that convention once, at the top of the schedule, rather than relying on a reader to infer it row by row. Where a schedule genuinely needs to mix conventions across sections, separate those sections visibly and label the convention change.